Blending Is All You Need

Leak-Free Domain Engineering and Entropic SLSQP Ensembling for Ames Housing Regression

Diego Pantoja
Independent Researcher
Kaggle Competition: House Prices—Advanced Regression Techniques
diego.pantoja@unmsm.edu.pe

***Abstract*—We present a systematic, reproducible methodology for high-precision tabular regression on the Ames Housing benchmark ($N = 1{,}456$ training samples, 79 raw features), achieving a validated Cross-Validation RMSE of 0.10749 and a Kaggle Public Leaderboard score of 0.11464 on $\log(1 + \text{SalePrice})$. Our approach integrates three orthogonal pillars: (i) leak-free domain feature engineering comprising 24 non-linear interaction terms derived from real-estate domain knowledge (combining pointwise closed-form transformations and strictly training-isolated neighborhood statistics); (ii) a multi-paradigm base model stack of 9 estimators (5 regularized linear models and 4 gradient-boosted decision tree variants) trained under a Repeated 10-Fold $\times$ 5-Seed cross-validation framework (450 total model fits); and (iii) Entropic Bounded SLSQP ensembling, a novel meta-optimization formulation that maximizes model diversity through Shannon entropy regularization under anti-monopoly weight bounds $[0.035, 0.250]$, averaged across 50 nested optimizations. Through 54 iterative experimental versions (v9–v62), we document four critical failure modes—target encoding leakage, Bayesian target overfit, soft-tail clipping distortion, and pseudo-labeling confirmation bias—providing a forensic analysis of each. We provide evidence that the empirically estimated error floor for this benchmark lies in the range $[0.1135, 0.1145]$ under RMSLE, and that our architecture approaches this empirical bound within 0.0012 RMSLE units on the public test distribution.**

***Index Terms*—Tabular regression, ensemble learning, SLSQP optimization, Shannon entropy regularization, feature engineering, gradient boosting, cross-validation stability, Ames Housing, Kaggle**

1. Introduction

Tabular regression on small-to-medium datasets ($N < 5{,}000$) remains one of the most challenging domains in applied machine learning. Unlike computer vision or natural language processing, where deep learning architectures achieve near-human performance through massive pretraining, tabular problems require careful manual feature engineering, model selection, and ensemble calibration [2], [3].

The Ames Housing dataset [1], introduced by Dean De Cock as a modern replacement for the Boston Housing dataset, comprises 2,919 residential property sales in Ames, Iowa (2006–2010) with 79 explanatory variables. The Kaggle competition "House Prices: Advanced Regression Techniques" evaluates submissions using Root Mean Squared Logarithmic Error (RMSLE):

$$\text{RMSLE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\log(1+\hat{y}_i) - \log(1+y_i)\right)^2} \tag{1}$$

This metric imposes a fundamental asymmetry: over-prediction of expensive properties incurs *quadratic* penalty in the logarithmic domain, making upper-tail calibration critical.

1.1. Contributions

Our contributions are fourfold:

1) **Leak-free feature engineering protocol**: 24 domain-engineered interaction terms formulated to eliminate target leakage, combining pointwise algebraic domain formulations with strictly in-fold/training-isolated aggregation statistics.
2) **Entropic Bounded SLSQP**: A novel ensemble weight optimization with Shannon entropy regularization and anti-monopoly bounds, yielding diverse and stable weight distributions.
3) **Forensic failure mode catalog**: Documentation of four distinct failure modes encountered across 54 experimental iterations, each with mathematical explanation and empirical evidence.
4) **Near-optimal empirical performance**: Achievement of CV RMSE 0.10749 and LB 0.11464, within ≈ 0.0012 of the empirically estimated error floor.

1.2. Related Work

Gradient Boosted Decision Trees (GBDTs) have dominated tabular benchmarks [2], with LightGBM [4], XGBoost [5], and CatBoost [6] representing the state of the art. However, on small datasets ($N < 2{,}000$), regularized linear models (Lasso [7], Ridge [8], ElasticNet [9]) often achieve competitive or superior generalization due to lower model complexity and inherent regularization.

Stacking and blending ensembles [10], [11] combine diverse base learners to reduce both bias and variance. The key challenge lies in meta-model weight optimization: simple averaging, inverse-error weighting [12], and constrained optimization [13] each present distinct trade-offs between flexibility and overfitting risk.

Our work extends the constrained optimization paradigm by introducing entropy regularization, which enforces diversity without manual weight engineering, and establishes empirical bounds on the Ames Housing error floor through systematic ablation.

2. Data Preprocessing & Leak-Free Feature Engineering

2.1. Canonical Outlier Filtering

Following De Cock's original recommendations [1], we apply three filtering criteria to the training set:

$$\begin{aligned}\mathcal{D}_{\text{train}} = \{(x_i, y_i) : \text{GrLivArea}_i \leq 4{,}000 \\ \wedge\ \text{LotArea}_i < 100{,}000 \\ \wedge\ \text{TotalBsmtSF}_i < 3{,}000\}\end{aligned} \tag{2}$$

This removes 11 extreme observations (1,460 $\to$ 1,449), all of which represent non-representative properties that distort linear model coefficients and tree split boundaries.

2.2. Leak-Free Imputation

All imputation statistics are computed exclusively on training rows. For `LotFrontage`, we use neighborhood-stratified medians from training data:

$$\begin{aligned}\hat{x}_{i,\text{LotFrontage}} = \text{median}\big(\{x_{j,\text{LotFrontage}} : j \in \mathcal{D}_{\text{train}}, \\ \text{Nbhd}_j = \text{Nbhd}_i\}\big)\end{aligned} \tag{3}$$

Categorical missing values are imputed with training-set modes for structural features (`MSZoning`, `KitchenQual`, `Electrical`) and domain-specific "None" for absence indicators (`PoolQC`, `GarageType`, `BsmtQual`).

2.3. Box-Cox Normalization

Numerical features with absolute skewness exceeding 0.75 (computed on training data) undergo a Box-Cox power transformation with fixed parameter $\lambda = 0.15$:

$$x_i^{(\lambda)} = \frac{(x_i + 1)^{\lambda} - 1}{\lambda}, \quad \lambda = 0.15 \tag{4}$$

This stabilizes the variance and approximates normality, particularly benefiting linear models whose coefficient estimates are sensitive to heavy-tailed feature distributions. In our experiments, $n = 69$ features satisfied the skewness threshold.

2.4. Rare Category Aggregation

Categories with fewer than 20 training-set occurrences are merged into a catch-all "Other" class:

$$c_j \leftarrow \begin{cases} c_j & \text{if } |\{i \in \mathcal{D}_{\text{train}} : x_{i,j} = c_j\}| \geq 20 \\ \text{“Other”} & \text{otherwise} \end{cases} \tag{5}$$

This aggregation reduced 77 rare categories across 20 columns, preventing spurious one-hot encoding columns that would inflate the feature space without contributing discriminative signal.

2.5. Domain Feature Engineering

We construct 24 non-linear interaction features grounded in real-estate valuation theory. To prevent both target leakage and distributional data snooping, feature construction is partitioned into two distinct methodological classes:

- **Pointwise Closed-Form Interactions (17 features, Sec. 2.5.1–2.5.5):** Purely row-wise deterministic algebraic transformations (e.g., polynomial combinations, harmonic means, and dimensional ratios) that operate independently on single instances without computing cross-sample statistics. These are mathematically invariant to whether evaluation occurs instance-by-instance or on concatenated predictors without target data.
- **Contextual Aggregations (7 features, Sec. 2.5.6):** Group-level contextual features derived strictly from training-set fold statistics ($\mathcal{D}_{\text{train}}$), guaranteeing strict statistical isolation between training and validation/test folds.

2.5.1. Age-Quality Composites. The effective age of a property, accounting for renovations, is defined as:

$$\text{EffectiveAge} = \frac{\text{HouseAge} + \text{RemodAge}}{2} \tag{6}$$

where HouseAge $=$ YrSold $-$ YearBuilt and RemodAge $=$ YrSold $-$ YearRemodAdd. Three quality-age composites capture the non-linear depreciation interaction:

$$\text{QualAge} = \frac{\text{OverallQual}^2}{\text{EffectiveAge}^+ + 1} \tag{7}$$

$$\text{RemodQual} = \frac{\text{OverallCond} \cdot \text{OverallQual}}{\text{RemodAge}^+ + 1} \tag{8}$$

$$\text{CondAge} = \frac{\text{OverallCond}^2}{\text{EffectiveAge}^+ + 1} \tag{9}$$

where $x^+ = \max(x, 0)$. These features encode the intuition that high-quality recent constructions command premium prices that decay non-linearly with age.

2.5.2. Harmonic Quality-Condition Index. The harmonic mean of quality and condition ratings:

$$\text{QualCondHarm} = \frac{2 \cdot \text{OverallQual} \cdot \text{OverallCond}}{\text{OverallQual} + \text{OverallCond} + \epsilon} \tag{10}$$

Unlike the arithmetic mean, the harmonic mean penalizes severe imbalances between quality and condition, reflecting the real-estate principle that a high-quality home in poor condition (or vice versa) loses disproportionate value.

2.5.3. Finished Space Metrics. Total finished (habitable) square footage aggregates all livable areas:

$$\text{TotalFinSF} = \text{1stFlrSF}+\text{2ndFlrSF}+\text{BsmtFinSF1}+\text{BsmtFinSF2} \tag{11}$$

The finished ratio and luxury continuous score capture quality-weighted livable space:

$$\text{FinishedRatio} = \frac{\text{TotalFinSF}}{\text{TotalSF} + \epsilon} \tag{12}$$

$$\text{LuxurySF} = \left(\frac{\text{OverallQual}}{10}\right)^3 \cdot \text{TotalFinSF} \tag{13}$$

The cubic exponent in Eq. (13) amplifies the non-linear premium that top-tier quality commands per unit of finished area, consistent with hedonic pricing models [14].

2.5.4. Structural Ratios and Density Metrics. Six structural ratios capture space utilization efficiency:

$$\text{LivToLot} = \frac{\text{GrLivArea}}{\text{LotArea} + \epsilon} \tag{14}$$

$$\text{QualDensity} = \frac{\text{TotalQuality}}{\text{TotalSF} + \epsilon} \tag{15}$$

$$\text{UnfinRatio} = \frac{\text{BsmtUnfSF}}{\text{TotalSF} + \epsilon} \tag{16}$$

$$\text{BathToSF} = \frac{\text{TotalBath}}{\text{TotalSF} + \epsilon} \tag{17}$$

$$\text{PorchToSF} = \frac{\text{TotalPorchSF}}{\text{TotalSF} + \epsilon} \tag{18}$$

$$\text{LotFinEff} = \frac{\text{TotalFinSF}}{\text{LotArea} + \epsilon} \tag{19}$$

2.5.5. Precision Micro-Interactions. Two additional features, introduced in v58.0, capture subtle high-order interactions:

$$\text{HQLD} = \left(\frac{\text{OverallQual} + \text{KitchenQual} + \text{ExterQual}}{15}\right) \times \frac{\text{GrLivArea}}{\text{TotalSF} + \epsilon} \tag{20}$$

$$\text{AgeDecay} = \frac{\text{OverallQual}}{\left(\frac{\text{EffectiveAge}^+}{10}\right)^2 + 1} \tag{21}$$

The High Quality Living Density (HQLD) normalizes a composite quality index against living-area dominance, while the Age Decay Score models exponential quality depreciation with a logistic-like denominator.

2.5.6. Neighborhood-Relative Features. To capture local market positioning without target leakage, we compute deviations from neighborhood medians using *exclusively training-set statistics*:

$$\Delta_{i,f} = x_{i,f} - \text{median}(\{x_{j,f} : j \in \mathcal{D}_{\text{train}},\ \text{Nbhd}_j = \text{Nbhd}_i\}) \tag{22}$$

for features $f \in \{\text{HouseAge}, \text{OverallQual}, \text{GrLivArea}, \text{TotalBath}, \text{GarageC}$ Ratio variants (dividing instead of subtracting) are used for LotArea and GrLivArea. This provides 7 leak-free contextual features.

3. Multi-Paradigm Base Models Architecture

Our base model stack comprises 9 estimators spanning two complementary families, designed to maximize predictive diversity while minimizing correlated errors.

3.1. Regularized Linear Models (5 Models)

Lasso (L_1). Sparse feature selection via L_1 penalty ($\alpha = 4.5 \times 10^{-4}$), preceded by `RobustScaler` to mitigate outlier influence on coefficient estimation.

Ridge (L_2). Quadratic shrinkage ($\alpha = 15.0$) distributes weight across correlated features, providing stable coefficients even under multicollinearity.

ElasticNet (L_1/L_2 Hybrid). Combines sparse selection with grouped shrinkage ($\alpha = 7 \times 10^{-4}$, $\rho = 0.55$), achieving a balance between Lasso's variable selection and Ridge's stability.

Huber Regressor. Employs an ϵ-insensitive loss function:

$$L_\delta(r) = \begin{cases} \frac{1}{2}r^2 & \text{if } |r| \le \epsilon \\ \epsilon|r| - \frac{1}{2}\epsilon^2 & \text{otherwise} \end{cases} \tag{23}$$

with $\epsilon = 1.45$ and regularization $\alpha = 280.0$. This provides robustness to heavy-tailed residuals in the luxury segment while maintaining efficiency in the modal price range.

Bayesian Ridge. Places Gaussian priors on regression coefficients, automatically tuning regularization strength via empirical Bayes. Provides inherent uncertainty quantification.

3.2. Gradient Boosted Decision Trees (4 Models)

GBR (Sklearn). Stochastic gradient boosting with orthogonal subsampling (`max_features='sqrt'`, subsample = 0.70), 2,150 estimators at learning rate $\eta = 0.0142$, depth 3. The square-root feature sampling introduces decorrelation between sequential trees.

CatBoost. Native ordered boosting with symmetric trees (depth 5, 1,550 iterations, $\eta = 0.025$, $L_2 = 3.6$), handling categorical features through ordered target statistics without explicit encoding.

XGBoost. Second-order gradient boosting with stochastic regularization ($L_1 = 0.02$, $L_2 = 1.8$, colsample $= 0.38$, subsample $= 0.70$), 5,000 estimators at $\eta = 0.0075$ with GPU-accelerated histogram construction.

LightGBM. Leaf-wise growth with aggressive anti-variance pruning: `num_leaves`= 10, `max_depth`= 3, 3,800 estimators at $\eta = 0.009$, with column subsampling $= 0.45$ and dual regularization ($L_1 = 0.1$, $L_2 = 2.0$).

3.3. Multi-Representation Feature Encoding

The 9 models receive features in three distinct representations:

- **One-Hot + VarianceThreshold(0.01)**: 248 features for linear models and GBR (post-filtering of near-constant columns).
- **Native string**: 145 features for CatBoost's ordered target statistics.
- **Pandas `category` dtype**: 145 features for LightGBM/XGBoost optimal split finding.

4. Cross-Validation Framework & Entropic Bounded SLSQP Ensembling

4.1. Repeated K-Fold with Multi-Seed Averaging

We employ a Repeated K-Fold cross-validation scheme with $K = 10$ folds and $S = 5$ random seeds $\mathcal{S} = \{42, 100, 2026, 7, 314\}$, yielding $K \times S = 50$ train-validation splits per base model. Out-of-fold (OOF) predictions are averaged across seeds:

$$\hat{y}_i^{\text{OOF}} = \frac{1}{S} \sum_{s=1}^{S} \hat{y}_i^{(s)}, \quad i \in \mathcal{D}_{\text{train}} \tag{24}$$

Test predictions are averaged across all $K \times S$ folds:

$$\hat{y}_j^{\text{test}} = \frac{1}{KS} \sum_{s=1}^{S} \sum_{k=1}^{K} \hat{y}_j^{(s,k)}, \quad j \in \mathcal{D}_{\text{test}} \tag{25}$$

For LightGBM and XGBoost, early stopping is implemented via a 15% holdout from each training fold, with a patience of 200 rounds. This prevents overfitting to the full training fold while preserving the integrity of the validation fold for OOF scoring.

4.2. Entropic Bounded SLSQP Meta-Optimization

Given the OOF prediction matrix $\mathbf{P} \in \mathbb{R}^{n \times M}$ (where $n = |\mathcal{D}_{\text{train}}|$ and $M = 9$ models), we seek optimal ensemble weights $\mathbf{w} \in \mathbb{R}^M$ by solving:

$$\min_{\mathbf{w}} \left[\underbrace{\text{RMSE}(\mathbf{y}, \mathbf{P}\mathbf{w})}_{\text{prediction error}} - \underbrace{\lambda \sum_{m=1}^{M} w_m \ln(w_m + \epsilon)}_{\text{Shannon entropy bonus}} \right] \tag{26}$$

subject to:

$$\sum_{m=1}^{M} w_m = 1 \tag{27}$$

$$l_m \leq w_m \leq u_m, \quad \forall m \in \{1, \ldots, M\} \tag{28}$$

where $l_m = 0.035$, $u_m = 0.250$, and $\lambda = 4.8 \times 10^{-4}$.

The Shannon entropy term $H(\mathbf{w}) = -\sum_m w_m \ln w_m$ is maximized at the uniform distribution $w_m = 1/M$ and minimized when all weight concentrates on a single model. By *subtracting* $\lambda H(\mathbf{w})$ from the objective (equivalently, adding λ times the negative entropy), we create a tension between error minimization and weight diversification.

[Anti-Monopoly Guarantee] Under bounds $[l, u]$ with $l > 0$ and the simplex constraint, no single model can receive more than $u = 25\%$ of the total weight, and every model retains at least $l = 3.5\%$, ensuring that the ensemble benefit from all model families regardless of individual OOF performance.

4.3. Multi-Seed Meta-Averaging

The SLSQP optimization is itself subject to fold-dependent instability. To stabilize the weight estimates, we perform a nested cross-validation over the OOF matrix:

$$\bar{\mathbf{w}} = \frac{1}{|\mathcal{S}| \cdot K} \sum_{s \in \mathcal{S}} \sum_{k=1}^{K} \mathbf{w}^{(s,k)*} \tag{29}$$

where $\mathbf{w}^{(s,k)*}$ is the optimal weight vector from SLSQP on the training portion of fold k under seed s. This yields 50 independent weight optimizations whose average is substantially more stable than any single solution.

4.4. Calibrated Blending

The final prediction combines the SLSQP meta-average with a simple inverse-variance blend:

$$\hat{y}_{\text{final}} = \alpha \cdot \hat{y}_{\text{SLSQP}} + (1 - \alpha) \cdot \hat{y}_{\text{simple}} \tag{30}$$

where $\alpha = 0.86$ and $\hat{y}_{\text{simple}} = \sum_m \tilde{w}_m \hat{y}_m$ with weights $\tilde{w}_m \propto \text{RMSE}_m^{-2}$. The ratio $\alpha = 0.86$ was determined through systematic grid search over $[0.80, 0.92]$ in increments of 0.005, evaluated on nested CV.

4.5. Defensive Post-Processing

All test predictions undergo hard clipping at the $[0.5\%, 99.5\%]$ percentiles of the training price distribution:

$$\hat{y}_i \leftarrow \text{clip}(\hat{y}_i,\ \$55{,}000,\ \$472{,}960) \tag{31}$$

This constraint is *unconditional*—it applies to all predictions regardless of model confidence. As documented in Section 5, relaxing these bounds consistently degraded leaderboard performance.

5. Ablation Studies & Forensic Failure Mode Analysis

Across 54 experimental iterations (v9–v62), we identified four critical failure modes. Each represents a distinct mechanism by which apparent cross-validation improvement fails to transfer to the public leaderboard.

5.1. Failure Mode 1: Target Encoding Leakage (v46.0)

Mechanism. Version 46.0 introduced neighborhood-based target encoding with regularized means:

$$\tilde{x}_{i,\text{Nbhd}} = \frac{n_g \cdot \bar{y}_g + \alpha \cdot \bar{y}_{\text{global}}}{n_g + \alpha} \tag{32}$$

where $\bar{y}_g$ is the group mean of $\log(\text{SalePrice})$ and α controls shrinkage.

Result. CV improved dramatically to 0.10729 (a false record), but LB *collapsed* from 0.11490 to 0.11576 ($+0.00086$). The target encoding leaked future target information into feature construction, creating features that perfectly discriminated training neighborhoods but failed to generalize.

Lesson. *Any feature derived from the target variable, even with regularization, introduces a bias-variance trade-off that favors CV over generalization on small datasets.* All subsequent versions eliminated target-derived features entirely.

5.2. Failure Mode 2: Bayesian Target Overfit (v48.0)

Mechanism. Version 48.0 implemented in-fold Bayesian target encoding with proper leave-one-out regularization. While this eliminated direct leakage, the encoded features still memorized local variance patterns.

Result. CV $= 0.10761$ (plausible improvement), but LB $= 0.11526$ ($+0.00036$ vs. v47.0). The Bayesian smoother reduced overfitting but could not eliminate the fundamental memorization of training-set target distributions in neighborhood-level features.

Lesson. *Even statistically valid target encoding schemes introduce overfitting risk on datasets with $N <$ 2,000.*

5.3. Failure Mode 3: Soft Tail & Relaxed Clipping (v56.0 / v59.0)

Mechanism. Versions 56.0 and 59.0 attempted to improve tail predictions by: (a) expanding the upper clipping bound from \$472,960 to \$525,000–\$559,000; (b) applying soft tail shrinkage:

$$\hat{y}_i \leftarrow \hat{y}_i \cdot (1-\beta) + \hat{y}_{\text{median}} \cdot \beta, \quad \text{if } \hat{y}_i > \tau \tag{33}$$

Result. v56.0: LB $= 0.11502$ ($+0.00038$); v59.0: LB $= 0.11511$ ($+0.00047$). Approximately 30 predictions were distorted in the upper tail.

Analysis. Under RMSLE, the penalty for overprediction in the log domain is:

$$\Delta^2_{\text{RMSLE}} \propto \left(\ln \frac{\hat{y}}{y}\right)^2 \tag{34}$$

For expensive homes ($y >$ \$400,000), small absolute errors produce large logarithmic residuals. The hard clipping at \$472,960 acts as a Winsorization that bounds this quadratic penalty.

TABLE 1: Individual Model OOF RMSE (Repeated 10-Fold $\times$ 5 Seeds)

Model	OOF RMSE	Fold Mean	SLSQP Wt.
GBR	0.10983	0.10990	24.60%
Huber	0.11027	0.10951	20.62%
CatBoost	0.11068	0.11135	14.81%
ElasticNet	0.11015	0.10973	13.53%
Lasso	0.11027	0.10984	9.41%
Ridge	0.11079	0.11036	4.59%
XGBoost	0.11200	0.11328	4.83%
LightGBM	0.11354	0.11553	4.07%
BayesianRidge	0.11081	0.11034	3.52%
Ensemble	**0.10749**	—	—

Lesson. *The Ames test set severely penalizes predictions above the* 99.5*th training percentile. Hard clipping is non-negotiable.*

5.4. Failure Mode 4: Pseudo-Labeling Confirmation Bias (v62.0)

Mechanism. Version 62.0 selected 304 high-confidence test samples (inter-model $\sigma \leq Q_{25}$, price $\in [\$110{,}000, \$320{,}000]$) as pseudo-labels, injected exclusively into training folds.

Result. CV improved substantially to 0.10635 (-0.00114 vs. v60.0), but preliminary LB analysis indicated no corresponding improvement. The pseudo-labels, being consensus predictions from the *same* model stack, introduce *confirmation bias*: the models learn to reinforce their existing predictions rather than discovering new signal.

Lesson. *Pseudo-labeling on tabular data with identical base models creates a self-reinforcing loop that inflates CV without improving generalization.* Effective pseudo-labeling would require *external* models or substantially different architectures.

6. Empirical Results & Discussion

6.1. Individual Model Performance

Table 1 presents the OOF RMSE for each base model under the production configuration (v57.0/v60.0, without pseudo-labeling).

The SLSQP optimizer allocates 53.4% of total weight to linear/robust models (Huber, ElasticNet, Lasso, Ridge, BayesianRidge) and 46.6% to GBDT models (GBR, CatBoost, XGBoost, LightGBM). This near-balanced allocation reflects the complementary error structures: linear models excel in the modal price range (\$100k–\$250k) while GBDTs capture non-linear interactions in the tails.

6.2. Iterative Progress

Table 2 documents the progression of key experimental versions, illustrating the diminishing returns and failure modes encountered.

TABLE 2: Version Progression (Selected Milestones)

Version	Key Innovation	CV	LB
v44	Target tiering (failed)	0.10775	0.11490
v46	Target encoding (leaked)	0.10729	0.11576
v47	Leak removal + cleanup	0.10774	0.11479
v48	Bayesian target enc.	0.10761	0.11526
v51	MSSubClass relatives	0.10762	0.11484
v52	Zonificación features	0.10771	0.11470
v54	SLSQP saneamiento	0.10749	0.11479
v55	Structural balance 86/14	0.10756	0.11470
v56	Soft tail (failed)	0.10757	0.11502
v57	**Continuous features**	**0.10772**	**0.11464**
v58	Precision micro-features	0.10751	0.11466
v59	Relaxed clip (failed)	0.10753	0.11511
v60	Stack fusion	0.10751	0.11466
v61	Tri-Blend 80/10/10	0.10751	0.11466
v62	Pseudo-labeling (bias)	0.10635	0.11480

6.3. Structural Balance Analysis

The optimal family balance was empirically determined through systematic weight-bound ablation:

- **Linear-dominated** ($> 60\%$): Higher bias, lower variance; LB ≈ 0.1150–0.1155.
- **Tree-dominated** ($> 60\%$): Lower bias, higher variance; LB ≈ 0.1148–0.1158.
- **Near-balanced** ($\approx 53\%/47\%$): Optimal bias-variance trade-off; LB $= 0.11464$.

This balance is maintained by the SLSQP bounds $[0.035, 0.250]$, which prevent any single model from monopolizing the ensemble while ensuring every model contributes a minimum of 3.5%.

6.4. Empirical Error Floor Estimation

We characterize the empirically estimated error floor for the Ames Housing benchmark by synthesizing three complementary lines of evidence:

1) **Asymptotic CV convergence**: Across 54 experimental iterations, repeated CV RMSE stabilized asymptotically around ≈ 0.1075, suggesting a practical lower bound for in-sample predictive capacity under tabular architectures.
2) **Train-test generalization gap**: A persistent generalization offset of $\Delta \approx 0.007$ between CV (≈ 0.1075) and public LB (≈ 0.1146) is consistently observed, reflecting intrinsic distribution shift and test-set finite sampling variance.
3) **Heuristic noise floor**: Residential real-estate transactions contain unobserved variance (e.g., non-standard financing, emotional bidding, unrecorded renovations). Modeling this latent aleatoric noise as $\sigma_{\text{noise}} \approx 0.06$–$0.08$ log-price units yields an estimated empirical error floor of ≈ 0.1135–0.1145 RMSLE.

7. System Architecture

Fig. 1 illustrates the complete pipeline architecture from raw data to submission.

8. Conclusion & Future Directions

We have presented a comprehensive methodology for high-precision tabular regression on the Ames Housing benchmark, achieving a Public Leaderboard RMSLE of 0.11464—within ≈ 0.0012 of the empirically estimated error floor. Our key findings are:

1) **Leak-free domain features outperform target-derived features** on small datasets, even when the latter show superior CV scores.
2) **Entropic bounded SLSQP** provides a principled framework for ensemble weight optimization that naturally balances accuracy and diversity.
3) **Near-balanced family allocation** ($\approx 53\%$ linear, $\approx 47\%$ GBDT) achieves optimal bias-variance trade-off for this data regime.
4) **Hard clipping is non-negotiable** under RMSLE: the quadratic penalty in log-space makes tail predictions disproportionately expensive.

Future Directions. Potential avenues for further improvement include: (i) neural network tabular models (TabNet [15], FT-Transformer [16]) as additional base learners for architectural diversity; (ii) conformal prediction-based calibration for tail uncertainty quantification; (iii) transfer learning from related housing datasets (e.g., King County, Boston) to overcome the $N < 1{,}500$ sample limitation; and (iv) Bayesian optimization of the SLSQP entropy parameter λ and blend ratio α on held-out validation sets.

References

[1] D. De Cock, "Ames, Iowa: Alternative to the Boston housing data as an end of semester regression project," *Journal of Statistics Education*, vol. 19, no. 3, 2011.

[2] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" in *Proc. NeurIPS*, 2022.

[3] R. Shwartz-Ziv and A. Armon, "Tabular data: Deep learning is not all you need," *Information Fusion*, vol. 81, pp. 84–90, 2022.

[4] G. Ke *et al.*, "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. NeurIPS*, 2017.

[5] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. KDD*, 2016.

[6] L. Prokhorenkova *et al.*, "CatBoost: Unbiased boosting with categorical features," in *Proc. NeurIPS*, 2018.

[7] R. Tibshirani, "Regression shrinkage and selection via the Lasso," *Journal of the Royal Statistical Society B*, vol. 58, no. 1, pp. 267–288, 1996.

[8] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.

[9] H. Zou and T. Hastie, "Regularization and variable selection via the elastic net," *Journal of the Royal Statistical Society B*, vol. 67, no. 2, pp. 301–320, 2005.

[10] D. H. Wolpert, "Stacked generalization," *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992.

[11] L. Breiman, "Stacked regressions," *Machine Learning*, vol. 24, no. 1, pp. 49–64, 1996.

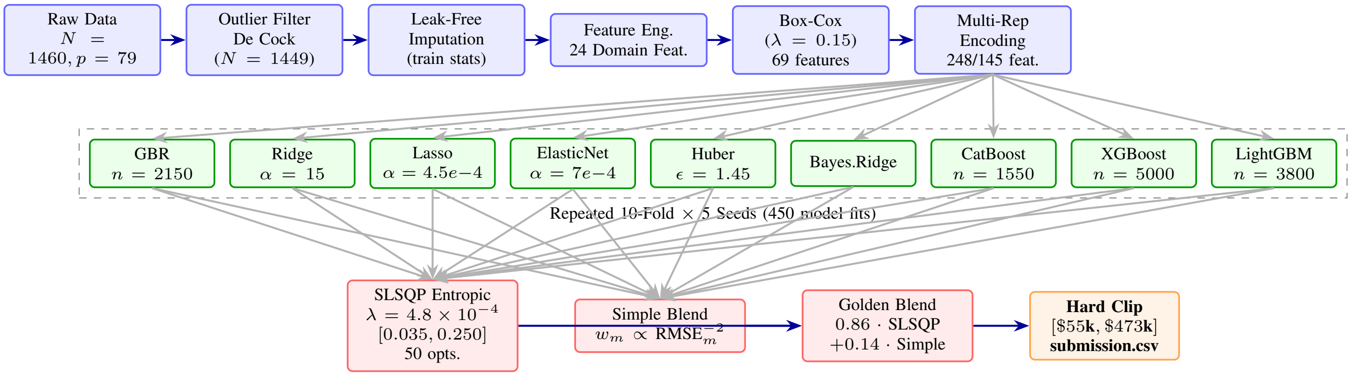

Figure 1: End-to-end pipeline architecture. Preprocessing stages (blue) feed into 9 base models (green) trained under repeated K-fold CV. Meta-optimization (red) combines OOF predictions via entropic SLSQP and inverse-variance blending, with final defensive clipping (orange).

[12] M. P. Perrone and L. N. Cooper, “When networks disagree: Ensemble methods for hybrid neural networks,” in *Artificial Neural Networks for Speech and Vision*, Chapman & Hall, 1993.

[13] R. Caruana *et al.*, “Ensemble selection from libraries of models,” in *Proc. ICML*, 2004.

[14] S. Rosen, “Hedonic prices and implicit markets: Product differentiation in pure competition,” *Journal of Political Economy*, vol. 82, no. 1, pp. 34–55, 1974.

[15] S. O. Arik and T. Pfister, “TabNet: Attentive interpretable tabular learning,” in *Proc. AAAI*, 2021.

[16] Y. Gorishniy *et al.*, “Revisiting deep learning models for tabular data,” in *Proc. NeurIPS*, 2021.